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Temporal disease dynamics and relative importance of sexual and asexual reproduction of grape downy mildew (*Plasmopara viticola*) in an isolated vineyard in the North Georgia Mountains, USA

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Abstract

The North Georgia Mountains are the southernmost region along the United States East Coast where European wine grapes (*Vitis vinifera*) are grown commercially. Epidemics of downy mildew, caused by *Plasmopara viticola*, are frequent and severe, but little is known about the epidemiology and population biology of the pathogen in this region. Disease monitoring in an experimental vineyard from 2015 to 2017 indicated that times of disease onset and progress rates were highly variable across years and cultivars. Oospores were observed microscopically, and simulation with a process-based model indicated presence of conditions favourable for oospore germination in the spring and early summer each year. A total of 409 *P. viticola* isolates collected over three years were genotyped with seven microsatellite markers, revealing very high genotypic diversity, which when combined with the observation of oospores is indicative of a sexually reproducing population. Among the 409 isolates, 225 multilocus genotypes (MLGs) were identified, of which 164 were detected only once and 61 were repeated (clonal). Eight MLGs (represented by 28 isolates) were detected across years, suggesting the possibility of asexual overwintering of *P. viticola* in this region. Across sampling dates, the percentage of isolates belonging to non-repeated (unique) MLGs ranged from 27.3% to 63.2%. Even towards the end of the annual epidemic, the percentage of isolates in nonrepeated MLGs was still relatively high, around 30%. These MLGs may have originated from oospores germinating late during the growing season, although incomplete sampling at earlier dates and contribution by immigration cannot be fully excluded.

KEYWORDS

downy mildew, grapevine, pathogen population genetics, *Plasmopara viticola*, *Vitis vinifera*

1 | INTRODUCTION

Grapes are currently produced on approximately 700 ha in the state of Georgia, USA (Wolfe and Stubbs, 2017), a small but rapidly expanding sector of the state's agricultural economy. The cumulative output of wineries, vineyards, and related sectors has been estimated to contribute up to $81.6 million to the economy statewide (Wolfe *et al.*, 2013). European wine grapes (*Vitis vinifera*), interspecific hybrids, and muscadine grapes (*V. rotundifolia*) are all used for wine production, but *V. vinifera* is largely restricted

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to the higher elevations (>400 m a.s.l.) in the Blue Ridge, Ridge and Valley, and Appalachian Plateau physiographic regions of northern Georgia.

Downy mildew, caused by *Plasmopara viticola*, is the most damaging disease of grapes in Georgia, causing an estimated 8% reduction in crop value in commercial conditions with active disease management (Brannen *et al*., 2017). The North Georgia Mountains are the southernmost region along the eastern United States where *V. vinifera* grapes are grown commercially. Little is known about the temporal dynamics of disease, the pathogen's mode of reproduction, and the relative importance of sexual and asexual inoculum in this emerging grape-producing environment. Previous work on identifying cryptic species and pairing isolates in controlled crosses on leaf disks has shown that *P. viticola* clade *aestivalis* (Pva) is the only cryptic species detected in the North Georgia Mountains and Piedmont area, where both mating types of Pva are also present (Hong *et al*., 2016, 2019). However, the presence of oospores and the roles of sexual and/or asexual reproduction during the growing season have not been documented. In some regions (e.g., on certain Greek islands and in Australia and South Africa), downy mildew epidemics are driven largely by asexual reproduction via sporangia during most of the growing season, resulting in pathogen populations dominated by few genotypes with large numbers of clones (Hug *et al*., 2006; Rumbou and Gessler, 2006; Koopman *et al*., 2007; Taylor *et al*., 2019). In contrast, oospores may germinate throughout the growing season in western Europe, contributing to a large proportion of infections being caused by nonrepeated (unique) genotypes (Gobbin *et al*., 2003a, 2005; Maddalena *et al*., 2020). Although it is typically assumed that *P. viticola* overwinters exclusively as oospores, there are a few studies inferring asexual survival based on microscopic observation (Barrett, 1939) or in situ detection of the same multilocus genotypes (MLGs) across years (Rumbou and Gessler, 2006; Koopman *et al*., 2007). Additionally, homothallic oospores have been hypothesized recently as a survival mechanism in unsuitable environments (Taylor *et al*., 2019).

Although North America is considered the centre of origin of *P. viticola*, detailed information regarding its epidemiology and population biology in eastern North America is limited (Rouxel *et al*., 2014), especially in the southeastern United States where most of the *V. vinifera* vineyards have been established relatively recently (within the past 20 years) and are small and geographically isolated. In this study, we monitored temporal patterns of downy mildew and genotypic composition of *P. viticola* populations at multiple time points over three years in a vineyard in the North Georgia Mountains to draw baseline inferences about the epidemiology of the disease and the mode of survival and reproduction of *P. viticola* in this region. We hypothesized that oospores are the only overwintering structure and initiate downy mildew epidemics in the spring. Thus, observation of oospores in infected leaves, high genotypic diversity early in the season, and a decreasing proportion of new MLGs (or increasing proportion of repeated MLGs) later in the season was expected. Furthermore, no repeated MLGs were expected across years if oospores are the only overwintering structures in this environment.

2 | MATERIALS AND METHODS

2.1 | Disease progress in the vineyard

Downy mildew was assessed at 4- to 15-day intervals in an experimental vineyard at the Georgia Mountain Research and Education Center (GMREC) in Blairsville, GA, USA from April to October in 2015, 2016, and 2017. Assessments were done visually and included presence of either signs (pathogen sporulation on foliage or grape berries) and/or symptoms (oil spots on leaves). The vineyard consisted of *V. vinifera* cultivars Merlot and Chardonnay, arranged in two contiguous blocks, with the vines being 6 years old at the initiation of the study. Seven rows of Merlot (56–66 vines per row) and eight rows of Chardonnay (51–54 vines per row) were included in 2015, but only five rows of Chardonnay were used in 2016 and 2017 due to weak growth of the plants in the other areas. Plant spacing was 1.8 m between vines and 3.6 m between rows.

Two treatments were implemented for each cultivar: (a) initially untreated with fungicide to determine the onset of the disease; these plots were switched to the "allow downy mildew" schedule after disease onset to maintain the overall health of the vines while allowing downy mildew to progress; and (b) "allow downy mildew" spray schedule consisting of fungicides with no or minimal effect on oomycetes but suppressing other diseases such as powdery mildew, black rot, and botrytis bunch rot (Table S1). Individual plots consisted of six or seven vines and were replicated four to six times in a randomized complete block design for each cultivar. Insecticide treatments, herbicide applications, and fertilization followed commercially used practices for wine grape production in the southeastern United States (Nita *et al*., 2016). Disease assessments were made across all plots in the vineyard.

Disease incidence was evaluated by assessing the proportion of leaves with *P. viticola* sporulation and/or oil spots on 108 arbitrarily selected fully expanded leaves per plot. At each assessment date, petioles of assessed leaves were marked with a marker pen, hence no leaves would be assessed repeatedly on the same date. Means were calculated and used to plot progress curves for disease incidence. Berry infection was assessed only in 2016, when 100 bunches in the "allow downy mildew" treatment were selected arbitrarily in late May and assessed in four plots per cultivar to calculate the disease incidence of clusters.

To compare epidemics among years and cultivars, standard growth models (monomolecular, logistic, and Gompertz) were fitted to the temporal disease progress data using PROC REG in SAS v. 9.4 (SAS Institute). Model fit was evaluated based on the maximum back-transformed R^2, minimum back-transformed mean square error (MSE), pattern of the residual plots, and the overlay plot of predicted against observed data. To compare apparent infection rates (r, with a unit of day^{-1}) among epidemics, r values estimated from different

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models were converted to weighted mean absolute rates (ρ) using the formula:

$$\rho = \frac{r}{2m+2}$$

where $m = 1$ when the Gompertz model best fitted the data and $m = 2$ when the logistic model best fitted the data.

2.2 | Assessment of oospore density

Leaves with chlorotic or necrotic mosaic lesions and/or presence of downy mildew sporulation on the lower leaf surface were collected from the canopy between September and October of 2015, 2016, and 2017. Leaves were air-dried in paper bags at ambient temperature and then crushed manually. In 2015 and 2016, 20–25 g of ground leaf material was covered with a nylon mesh, placed on 3–4 cm of sand in a plastic net pot (7.6 cm diameter) and incubated on the vineyard floor for at least 4 months, simulating the overwintering of oospores. Leaf material was subsequently retrieved to the laboratory, mixed, and stored at −20°C prior to oospore density assessment. In 2017, the leaves collected in the autumn were air-dried, ground, and then stored directly at −20°C prior to assessment. Oospore density was determined microscopically using a procedure modified from Spring and Zipper (2000) and Gent *et al*. (2017). Briefly, 1 g of leaf material was blended in a small stirrer (T-line 101; Talboys Engineering) in 25 ml deionized water. The tissue suspension was sonicated in an ultrasonic bath (Bransonic B-2200R-1; Branson) with ice for 15 min and then filtered through standard sieves with mesh sizes of 500 and 63 μm. The filtrate was centrifuged at 1,000 × *g* for 3 min and the pellet reconstituted in 1 ml deionized water. The tissue suspension was stained with acid fuchsin and observed microscopically to estimate the density of oospores. Three grams of leaf sample per year and three droplets of 10 μl suspension per sample were assessed.

2.3 | Modelling of oospore germination

The web-based Università Cattolica del Sacro Cuore (UCSC) downy mildew model was used to simulate oospore germination of *P. viticola* between mid-March and July in 2015, 2016, and 2017 (Rossi *et al*., 2008a, 2008b; Caffi *et al*., 2009). We only used the oospore germination module of the model, which simulates the number of germinating oospore cohorts over time. Each cohort indicates a relative proportion of oospores that was simulated to germinate on the same date based on hourly air temperature and rainfall data collected with a standard agrometeorological weather station at the GMREC.

2.4 | Isolate collection and DNA extraction

Downy mildew samples were collected at different times during disease progression and genotyped to estimate the relative contribution of sexually derived or asexually derived individuals to the epidemic; one sampling date in 2015 (22 July), four sampling dates in 2016 (27 May, 3 and 7 June pooled, 1 July, 13 August) and three sampling dates in 2017 (7 June, 27 June, 31 July). Samples (19 to 133 isolates for each date) consisted of individual leaves or berries (June 2016 only) with symptoms with single, actively sporulating downy mildew lesions collected from the canopy within 0.5 m of the trunk of a vine and 9–11 vines apart from the next sampled vine to cover the vineyard. Samples were returned to the laboratory in individual zipper storage bags, where an actively sporulating single colony of *P. viticola* was washed in 0.1 ml sterile deionized water and 10 μl of sporangial suspension was dispensed on one Chardonnay leaf disk (12 mm diameter) on rifampicin–ampicillin–pimaricin water agar in Petri dishes (Wong and Wilcox, 2000), 5–10 leaf disks per isolate in one dish. Dishes were incubated in the dark at 20°C for 16–24 hr, after which excessive water was removed by tapping the dishes upside down. Dishes were then incubated under 12 hr light at 20°C for 6–14 days, after which actively sporulating colonies were washed in 50 μl Tris-borate-EDTA buffer in a 1.5 ml microcentrifuge tube, microwaved for 20 s three times, and centrifuged at 12,735 × g for 5 min. Microsatellite PCR is robust (Gobbin *et al*., 2003b; Yin *et al*., 2010) and the DNA quality obtained from this method is sufficient for genotyping. The supernatant was stored at −20°C until use as template DNA for microsatellite genotyping.

2.5 | Genotyping with microsatellite markers

DNA of *P. viticola* isolates was PCR-amplified with seven microsatellite markers that have been used widely in population studies of *P. viticola* (Rouxel *et al*., 2012): Pv 61, Pv 83, Pv 88, Pv 91, Pv 93, Pv 101, and Pv 103. Forward primers of Pv 61 and Pv101 were labelled with 6FAM; Pv83 and Pv93 were labelled with NED; Pv 88 and Pv 103 were labelled with VIC; and Pv 91 was labelled with PET for multiplex PCRs. Two sets of primers, one for Pv61, Pv91, and Pv101, and the other for Pv 83, Pv88, Pv93, and Pv103 were multiplex PCR-amplified. Each set of reactions contained 1 μl of DNA template, 5 μl of 2 × Type-it Multiplex PCR mix (Qiagen), 0.2 μl of each forward and reverse primer mixture for one locus (10 μM), and RNase-free water added to a final volume of 10 μl. The PCR was conducted with an initial denaturation temperature of 94°C for 4 min; 38 cycles of 94°C for 30 s, 54°C for 30 s, and 72°C for 35 s; and final extension at 72°C for 5 min. PCR products of two multiplex PCR sets were combined and diluted 1:15 with ddH_2O, and 1 μl of the diluted product was added to a mixture of 9.9 μl Hi-Di formamide (Life Technologies) and 0.1 μl GeneScan-500 LIZ internal size standard (Applied Biosystems) in 96-well plates. The plates were heated at 95°C for 5 min and then cooled on ice before submitting to the Georgia Genomics and Bioinformatics Core (Athens, GA) for fragment analysis. Alleles of each locus were called based on the different fluorescent dyes and the size of alleles visualized in Geneious v. 10 (Kearse *et al*., 2012).

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2.6 | Population genetic analyses

The number of alleles and allelic richness among the 409 isolates were evaluated using HP-RARE 1.0 (Kalinowski, 2004, 2005). Observed and expected heterozygosity were calculated in GenAlEx v. 6.5 (Peakall and Smouse, 2006, 2012). MLGs were assigned and genotypic diversity (Simpson's index, λ) was estimated for each temporal subpopulation using poppr (Kamvar *et al*., 2014). P_{sex} values were estimated with the F_{IS} model and 1,000 simulations in MLGsim 2.0 to confirm that repeated MLGs were not the result of sexual reproduction (Stenberg *et al*., 2003). To determine the degree to which the subpopulations collected at the eight sampling dates were related, pairwise F_{ST} and associated significance levels were estimated based on complete and clone-corrected datasets with 1,000 permutations using GenAlEx v. 6.5. A Bonferroni correction was applied to correct significance levels for multiple comparisons.

3 | RESULTS

3.1 | Disease progress in the vineyard

For each of the 3 years, disease onset, which was defined as the first observation of downy mildew symptoms or signs on leaves, occurred between 2 and 3 weeks earlier on Chardonnay (between 11 May and 15 June) than on Merlot (24 May to 8 July; Figure 1). Subsequently, disease incidence (percentage of affected leaves among total leaves assessed) increased rapidly, reaching a proportion of >0.9 between 7 July and 25 August on the two cultivars. Leaves turned necrotic and vines began to defoliate during the period of fruit development through to beginning of grape ripening (BBCH 79–89) between mid-July and mid-August. Vines were completely defoliated due to the combination of downy mildew and cold weather by late October to early November (data not shown).

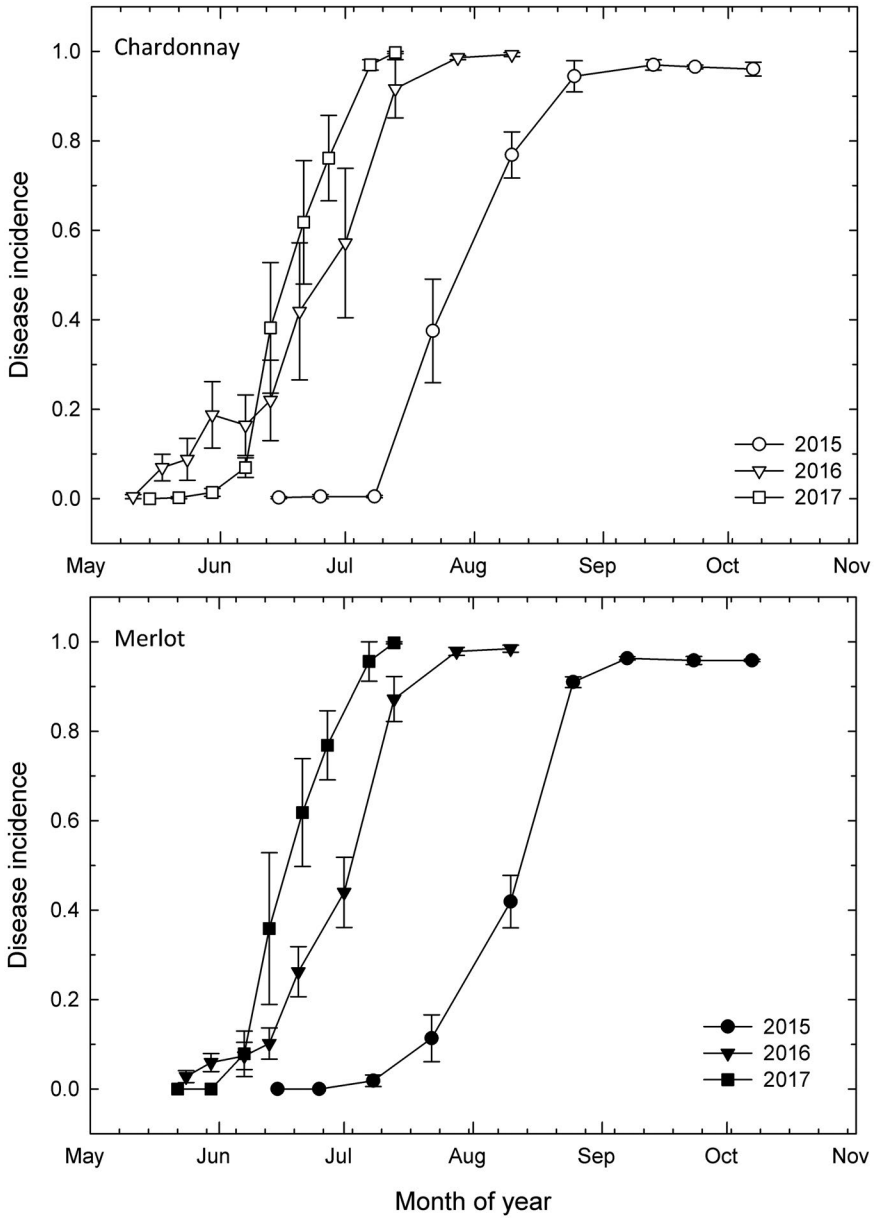


FIGURE 1 Disease progress of downy mildew on leaves of Chardonnay and Merlot vines in a vineyard in the Georgia Mountains (USA) in 2015, 2016, and 2017. Values are means and standard errors of four to six replicate plots

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The observed time from disease onset to disease incidence reaching a proportion of >0.9 was generally shorter on Merlot (mean 48 days) than on Chardonnay (60 days), which is consistent with the higher disease progress rate on Merlot (Table S2). Based on the simple growth models, the weighted mean absolute rates (ρ) for disease progress ranged from 0.0156 to 0.0393/day for Chardonnay and from 0.0179 to 0.0422/day for Merlot. Disease progress was fastest on both cultivars in 2017 (Figure 1; Table S2), which was also the year with the highest cumulative rainfall between April and August (69.9 mm compared with 62.3 mm in 2015 and 37.3 mm in 2016).

Symptoms and signs of berry infection by *P. viticola* (maximum disease incidence proportion between 0.30 and 0.43) were observed on both cultivars in 2016 and 2017 but not in 2015. The delayed downy mildew onset in 2015 (at the pea-sized berry stage of development, BBCH 75) may have occurred after the berries became ontogenetically resistant to *P. viticola* (Kennelly *et al*., 2005), resulting in the absence of berry infection.

3.2 | Oospores

Microscopic observation of leaves collected in the vineyard in the autumn of 2015, 2016, and 2017 and stored for various periods of time revealed the presence of oospores at densities of 811 ± 184, 789 ± 208, and 844 ± 249 per g of leaves (means ± *SE* of three samples), respectively (Figure 2). Because downy mildew-infected leaves were collected from the canopy and there is no report of other oomycetes causing diseases on grape leaves, it is unlikely that oospores observed in leaf material were produced by other oomycetes.

3.3 | Population genetic analyses

A total of 409 *P. viticola* isolates collected across eight sampling dates during the epidemics in 2015 (one sampling date), 2016 (four dates), and 2017 (three dates) were genotyped using the seven microsatellite markers. Preliminary analyses indicated that these markers should have sufficient power to resolve MLGs in the population (Figure S1). The overall genotypic diversity as measured with Simpson's index was very high (λ = 0.987 and 0.995 in the overall population for the complete and clone-corrected data set, respectively; Table 1).

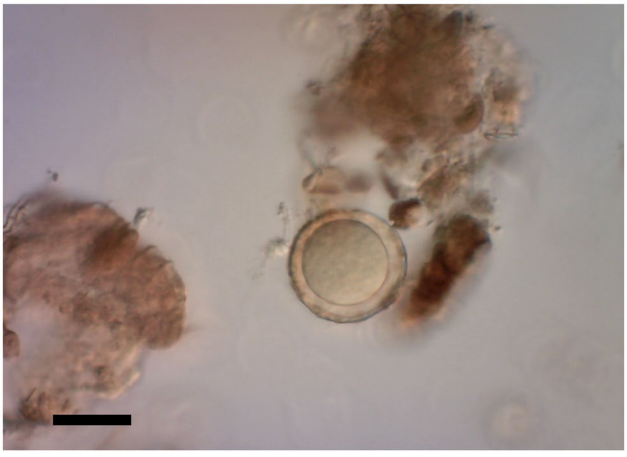


FIGURE 2 Oospore of *Plasmopara viticola* in *Vitis vinifera* leaves in the vineyard in the Georgia Mountains (USA). Bar = 20 μm [Colour figure can be viewed at wileyonlinelibrary.com]

For the overall population, a total of 22 alleles were detected, whilst the number of alleles in each temporal subpopulation varied from 15 to 19, with relatively constant allele composition and allele frequencies throughout the three years. In general, there were no consistent trends in any of the population genetic parameters over time within or across seasons. For example, λ for the complete data set increased from 0.943 to 0.973 in 2016 but decreased from 0.959 to 0.901 in 2017 across sampling dates (Table 1). Pairwise comparison of population differentiation among the eight temporal subpopulations showed low F_{ST} values (at or below 0.1 in all cases) indicating a general lack of population subdivision, and only two of the 28 subpopulation comparisons were significant at α = 0.05 for clone-corrected data (Table 2). This pattern again suggests that the temporal subpopulations in this vineyard had limited differentiation over time, and most of the inoculum source might have been coming from the oospores produced in previous seasons in the vineyard.

Among the 409 isolates in the total population, 225 MLGs were identified, of which 164 were detected only once and comprised 40.1% of the total population. Only 61 repeated MLGs were observed more than once within or across temporal populations, comprising 59.9% (245 isolates) of the total population (Table 3). All repeated MLGs had significant ($p < .1$) P_{sex} values (data not shown), indicating that they probably did not arise from sexual reproduction (Stenberg *et al*., 2003) and can be considered the result of clonal reproduction.

Across the eight temporal subpopulations, the percentage of isolates belonging to repeated MLGs ranged from 36.8% to 72.7% (Figure 3). Whereas the proportion of clones increased from earlier to later sampling dates in 2017, there was no such trend in 2016. Even on the last sampling dates toward the end of the 2016 and 2017 epidemics (13 August in 2016 and 31 July in 2017), the percentage of isolates belonging to nonrepeated MLGs was still relatively high at 39.8% and 27.3%, respectively. These MLGs may have originated from sexual recombination (oospores germinating later during the growing season), although immigration from outside the vineyard or incomplete sampling (i.e., the MLG was present at an earlier sampling date but was not detected) cannot be fully excluded.

4 | DISCUSSION

Downy mildew onset observed in this study was generally later on Merlot relative to Chardonnay, but the disease progressed faster once it occurred on Merlot. This trend is evident when comparing the weighted mean absolute rate (ρ) of the epidemics on both cultivars. In the literature, both cultivars are considered highly

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TABLE 1 Number of isolates (*N*), number of alleles (N_a), allelic richness (AR), number of multilocus genotypes (*Ng*), observed heterozygosity (H_o), unbiased expected heterozygosity (uH_e), and Simpson's index of genotypic diversity (λ) of temporal subpopulations of *Plasmopara viticola* from Chardonnay and Merlot vines in a vineyard in the Georgia Mountains (USA) across eight sampling dates in 2015, 2016, and 2017

Population[a]	*N*	N_a	AR	*Ng*	H_o	uH_e	λ
Jul_15	37	19	18.5 (17.6)	22	0.409 (0.364)	0.473 (0.486)	0.919 (0.955)
May_16	71	18	16.9 (16.2)	46	0.509 (0.438)	0.437 (0.445)	0.943 (0.978)
Jun_16	19	17	17.0 (16.9)	16	0.444 (0.411)	0.460 (0.472)	0.947 (0.938)
Jul_16	63	19	17.8 (17.1)	31	0.483 (0.378)	0.438 (0.452)	0.950 (0.968)
Aug_16	133	19	17.3 (17.1)	64	0.456 (0.451)	0.431 (0.467)	0.973 (0.984)
7Jun_17	28	17	16.8 (15.9)	26	0.337 (0.335)	0.406 (0.413)	0.959 (0.962)
27Jun_17	36	15	15.0 (15.0)	19	0.337 (0.286)	0.421 (0.461)	0.927 (0.947)
Jul_17	22	15	15.0 (15.0)	9	0.442 (0.429)	0.434 (0.458)	0.901 (0.889)
Total	409	22	–	233	0.427 (0.386)	0.438 (0.457)	0.987 (0.995)

Note: Numbers in parentheses correspond to clone-corrected data.

[a]All subpopulations were obtained from single leaf lesions in the month and year indicated, except subpopulation Jun_16, which was collected from infected berries.

TABLE 2 Pairwise comparison of differentiation among temporal subpopulations of *Plasmopara viticola* from Chardonnay and Merlot vines in a vineyard in the Georgia Mountains (USA) across eight sampling dates in 2015, 2016, and 2017

	Jul_15	May_16	Jun_16	Jul_16	Aug_16	7Jun_17	27Jun_17	Jul_17
Jul_15	–	0.045*	0.037	0.054*	0.095*	0.100*	0.052*	0.051
May_16	0.028	–	0.004	0.004	0.026*	0.035	0.014	0.029
Jun_16	0.015	0.004	–	0.000	0.005	0.031	0.020	0.010
Jul_16	0.036	0.006	0.000	–	0.007	0.016	0.022	0.013
Aug_16	0.036	0.030*	0.000	0.011	–	0.017	0.050*	0.028
7Jun_17	0.093*	0.036	0.018	0.002	0.035	–	0.023	0.003
27Jun_17	0.029	0.000	0.000	0.000	0.016	0.004	–	0.018
Jul_17	0.041	0.025	0.000	0.000	0.035	0.007	0.000	–

Note: F_{ST} values based on complete data set shown above the diagonal, whereas those based on the clone-corrected data set are shown below. Asterisks (*) denote significance at $p < .00179$ after Bonferroni correction.

susceptible to downy mildew (Babadoost, 2004). The differences between them observed in the present study could have been due to the relative location of the two cultivar blocks in the vineyard, where the Merlot section was more shaded by surrounding woodlands than the Chardonnay section.

Delayed disease onset on both cultivars was observed in 2015, which may have been due in part to an application of mancozeb fungicide that occurred at budbreak in 2015 but not in 2016 or 2017. Mancozeb is reported to have some protective activity against downy mildew (Wong and Wilcox, 2001), which may have delayed disease onset until the chemical was washed off by rain or degraded.

Previous research documented the presence of both mating types of *P. viticola* in vineyards in the North Georgia Mountains (Hong *et al*., 2016); however, this is the first report of the occurrence of oospores of the pathogen in this grape production region. Downy mildew oospores in a vineyard consist of different age cohorts that may mature and germinate at different times over the growing season (Jermini *et al*., 2003; Rossi *et al*., 2008b; Vercesi *et al*., 2010). In the UCSC model, precipitation >0.2 mm/hr or a vapour pressure deficit of ≤4.5 hPa at 20°C are considered to initiate germination of a cohort of physiological mature oospores (Rossi *et al*., 2008a, 2008b). Although a floating disk baiting method (Caffi *et al*., 2011) applied to the overwintered leaf samples in our study showed only a very low incidence of oosporic infection in preliminary tests (data not shown), weather-based simulation of oospore development with the UCSC downy mildew model predicted germination of oospores between mid-March and early July of 2015, 2016, and 2017 (38, 32, and 54 cohorts of germinating oospores, respectively), suggesting that environmental conditions at the site would have been favourable for oospore germination and subsequent development of downy mildew.

In addition to the physical presence of oospores in leaf litter and their predicted germination with the downy mildew model, the results of our population genetic analyses also supported a role of oospores in the disease cycle at the study site. Observed genotypic diversity was very high, and within a given year where multiple sampling dates were available (2016 and 2017), the proportion of new

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TABLE 3 Number of *Plasmopara viticola* isolates from Chardonnay and Merlot vines belonging to repeated (clonal) or nonrepeated (unique) multilocus genotypes (MLGs) in a vineyard in the Georgia Mountains (USA) across eight sampling dates in 2015, 2016, and 2017

Population[a]		Jul_15	May_16	Jun_16	Jul_16	Aug_16	7Jun_17	27Jun_17	Jul_17
Nonrepeated MLGs		16	27	12	24	53	13	13	6
Repeated	MLG190	7							
	MLG121	5							
	MLG105	3							
	MLG20	2							
	MLG187	2							
	MLG48	2	2			1			
	MLG161		12		6	4			
	MLG152		9	1	10	2			
	MLG31		2		3				
	MLG158		2		1	1			
	MLG39		2						
	MLG97		2						
	MLG135		2						
	MLG153		1	1					
	MLG61		1	1		4			
	MLG100		1		2				
	MLG113		1		1				
	MLG159		1		1				
	MLG162		1		1				
	MLG147		1		2	8			
	MLG122		1			2			
	MLG69		1			1			
	MLG169		1			1			
	MLG63		1			2		1	
	MLG15			1	1				
	MLG141			1	1				
	MLG51			1		5			
	MLG143			1		2			
	MLG64				2				
	MLG186				2				
	MLG57				2	4		1	
	MLG88				1	2			
	MLG165				1	1			
	MLG213				1	1			
	MLG96				1			1	
	MLG154					12			
	MLG54					8			
	MLG150					5			
	MLG145					4			
	MLG104					2			
	MLG109					2			
	MLG142					2			
	MLG66					1	1		

(Continues)

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TABLE 3 (Continued)

Population[a]		Jul_15	May_16	Jun_16	Jul_16	Aug_16	7Jun_17	27Jun_17	Jul_17
	MLG68					1	1		
	MLG164					1	2		
	MLG11					1			2
	MLG75						2		
	MLG225						1	7	
	MLG76						1	1	
	MLG95						1	1	
	MLG192						1	1	
	MLG206						1	1	
	MLG149						1	4	5
	MLG36						1		1
	MLG58						1		1
	MLG200						1		1
	MLG14							2	1
	MLG201							2	
	MLG219							1	1
	MLG124								2
	MLG9								2

[a]All subpopulations were obtained from single leaf lesions in the month and year indicated, except subpopulation Jun_16 which was collected from infected berries. Shaded MLGs were detected across multiple years.

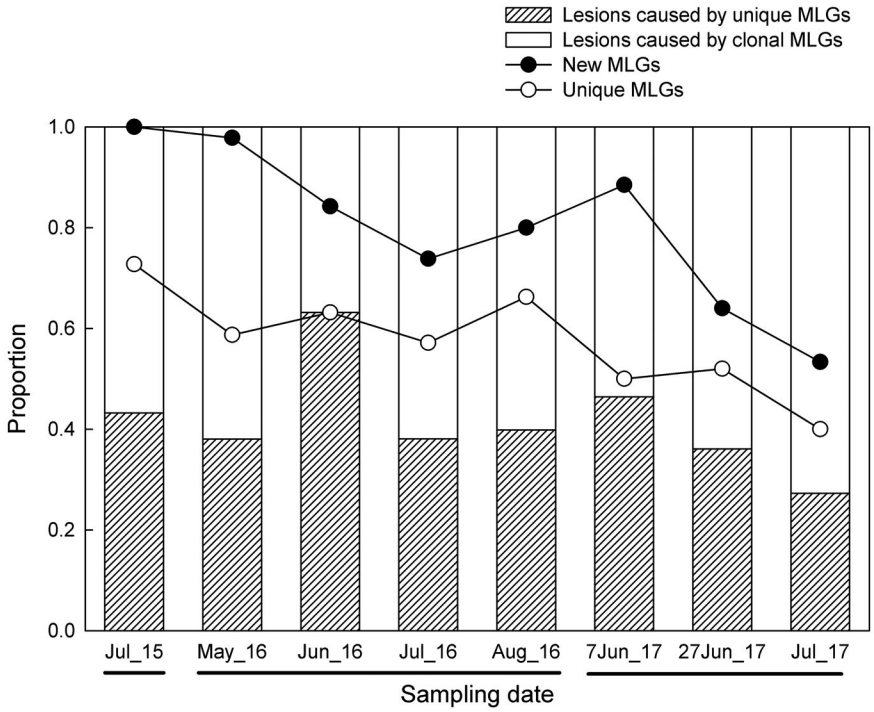


FIGURE 3 Proportion of repeated (clonal) and nonrepeated (unique) multilocus genotypes (MLGs) of *Plasmopara viticola* from Chardonnay and Merlot vines in a vineyard in the Georgia Mountains (USA) across eight sampling dates in 2015, 2016, and 2017. Also shown is the proportion of new MLGs detected for the first time in each temporal subpopulation and nonrepeated MLGs in all populations relative to all MLGs recorded cumulatively up to that sampling date

MLGs was always highest at the first sampling date, which may reflect the effect of sexual recombination on the population following overwintering by oospores. In addition, across the eight sampling dates, the proportion of MLGs detected for the first time (new MLGs) in that temporal subpopulation relative to all MLGs recorded cumulatively up to that sampling date showed a decreasing trend. This decreasing proportion of new MLGs is consistent with previous studies that proposed that different age cohorts of oospores mature and germinate not only in the spring, but also throughout the summer, contributing new genetic variants throughout the growing season, albeit at a decreasing level over time (Jermini *et al*., 2003; Kennelly *et al*., 2004; Rossi *et al*., 2008b; Vercesi *et al*., 2010).

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With regard to asexual reproduction, only 61 clonal MLGs were found to be involved in secondary infection in the present study, but these clonal MLGs represented 59.9% of the collected lesions. Most clonal MLGs produced fewer than five clones and only a few produced more than 10 clones. Variation in asexual propagation during an epidemic among MLGs has been documented in other studies; in some cases, a few clonal MLGs dominated the epidemic by producing more than 90% of detected lesions, whereas in other cases most clonal MLGs contributed no more than 10% of lesions (Rumbou and Gessler, 2004; Gobbin *et al*., 2005; Koopman *et al*., 2007). Whether MLGs with more clones were simply established earlier in the vineyard or whether they have more favourable fitness traits requires further study.

In this study, 40 repeated MLGs represented by 173 of the 409 isolates were observed on more than one sampling date, of which 8 MLGs represented by 28 isolates were detected across years. The latter observation suggests asexual overwintering of *P. viticola* from one year to the next in this vineyard. Asexual overwintering of *P. viticola* has not been studied extensively, although there are several reports suggestive of this phenomenon. For example, the mycelium of *P. viticola* was reported to overwinter in cortical tissues and cane buds, and could be observed microscopically in parenchyma tissues of *Vitis californica* in California (Barrett, 1939). Recent studies based on microsatellite markers consistently detected the same MLG across years in grape-producing regions of South Africa and on a Greek island, suggesting asexual overwintering of the pathogen (Rumbou and Gessler, 2006; Koopman *et al*., 2007). In addition, formation of homothallic oospores was also proposed as a potential mechanism for surviving in unsuitable environments (Taylor *et al*., 2019). The potential for overwintering or survival of *P. viticola* as asexual structure(s), such as mycelium in host tissues, or the possibility of forming selfed oospores in field conditions needs to be studied by combining different strategies to provide more conclusive evidence for these hypotheses.

In conclusion, downy mildew progresses very rapidly in the humid *V. vinifera* production region in the North Georgia Mountains, although the time of disease onset can be highly variable. The pathogen produces oospores, but limited asexual overwintering may be possible as well, based on the presence of the same MLGs detected across years. The pathogen population within this vineyard is highly diverse, and unique MLGs continue to appear later in the growing season, suggesting that oospores may germinate and contribute inoculum beyond the spring. Although rapid disease progression and high genetic plasticity present significant challenges for practical disease management, strategies focusing on reducing the contribution of primary inoculum throughout the growing seasons may warrant further evaluation for slowing down disease progression.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

Babadoost, M. (2004) Report on plant disease – Downy mildew of grape. RPD 705 University of Illinois Extension, Urbana-Champain, IL. Available at: https://ipm.illinois.edu/diseases/rpds/705.pdf [Accessed 12 August 2020]

Barrett, J.T. (1939) Overwintering mycelium of *Plasmopara viticola* (B & C) Berl. & Det. in the California wild grape, *Vitis californica* Benth. *Phytopathology*, 29, 822–823.

Brannen, P.M., Brock, J., Dutta, B., Jagdale, G., Jogi, A., Kemerait, R. & *et al*. (2017) Bunch grape. In: *2015 Georgia Plant Disease Loss Estimates*. Athens, GA: University of Georgia Cooperative Extension.

Caffi, T., Rossi, V., Bugiani, R., Spanna, F., Flamini, L., Cossu, A. *et al*. (2009) A model predicting primary infections of *Plasmopara viticola* in different grapevine-growing areas of Italy. *Journal of Plant Pathology*, 91, 535–548.

Caffi, T., Rossi, V. and Lusitani, M. (2011) Long-term survival of *Plasmopara viticola* oospores. *IOBC/WPRS Bulletin*, 67, 111–114.

Gent, D.H., Cohen, Y. and Runge, F. (2017) Homothallism in *Pseudoperonospora humuli*. *Plant Pathology*, 66, 1508–1516.

Gobbin, D., Pertot, I. and Gessler, C. (2003a) Genetic structure of a *Plasmopara viticola* population in an isolated Italian mountain vineyard. *Journal of Phytopathology*, 151, 636–646.

Gobbin, D., Pertot, I. and Gessler, C. (2003b) Identification of microsatellite markers for *Plasmopara viticola* and establishment of high throughput method for SSR analysis. *European Journal of Plant Pathology*, 109, 153–164.

Gobbin, D., Jermini, M., Loskill, B., Pertot, I., Raynal, M. and Gessler, C. (2005) Importance of secondary inoculum of *Plasmopara viticola* to epidemics of grapevine downy mildew. *Plant Pathology*, 54, 522–534.

Hong, C., Brannen, P.M. and Scherm, H. (2016) Epidemiology and population biology of grape downy mildew (*Plasmopara viticola*) in Georgia. *Phytopathology*, 106(S4), 11.

Hong, C., Brewer, M.T., Brannen, P.M. and Scherm, H. (2019) Prevalence, geographic distribution and phylogenetic relationships among cryptic species of *Plasmopara viticola* in grape-producing regions of Georgia and Florida, USA. *Journal of Phytopathology*, 167, 422–429.

Hug, F., Gobbin, D., Gesseler, C. and Magarey, P. (2006) Genetic structure and epidemiology of *Plasmopara viticola* populations from Australian grape growing regions. In: Pertot, I., Gesseler, C., Gadoury, D., Gubler, W., Kassemeyer, H.H. and Magarey, P. (Eds.) *Proceedings of the 5th International Workshop on Grapevine Downy and Powdery Mildew*. San Michele all'Adige, Italy: Istituto Agrario di San Michele all'Adige, pp. 64–65.

Jermini, M., Gobbin, D., Blaise, P. and Gessler, C. (2003) Influence of the overwintering methods on the germination dynamic of downy mildew (*Plasmopara viticola*) oospores. *IOBC/WPRS Bulletin*, 26, 37–41.

13653059, 2020, 9, Downloaded from https://bsppjournals.onlinelibrary.wiley.com/doi/10.1111/ppa.13263 by Readcube (Labtiva Inc.), Wiley Online Library on [24/06/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kalinowski, S.T. (2004) Counting alleles with rarefaction: Private alleles and hierarchical sampling designs. *Conservation Genetics*, 5, 539–543.
Kalinowski, S.T. (2005) HP-RARE 1.0: a computer program for performing rarefaction on measures of allelic richness. *Molecular Ecology Notes*, 5, 187–189.
Kamvar, Z.N., Tabima, J.F. and Grünwald, N.J. (2014) Poppr: an R package for genetic analysis of populations with clonal, partially clonal, and/or sexual reproduction. *PeerJ*, 2, e281.
Kearse, M., Moir, R., Wilson, A., Stones-Havas, S., Cheung, M., Sturrock, S. *et al.* (2012) Geneious Basic: an integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics*, 28, 1647–1649.
Kennelly, M.M., Eugster, C., Gadoury, D.M., Smart, C.D., Seem, R.C., Gobbin, D. *et al.* (2004) Contributions of oosporic inoculum to epidemics of grapevine downy mildew (*Plasmopara viticola*). *Phytopathology*, 94, S50–S51.
Kennelly, M.M., Gadoury, D.M., Wilcox, W.F., Magarey, P.A. and Seem, R.C. (2005) Seasonal development of ontogenic resistance to downy mildew in grape berries and rachises. *Phytopathology*, 95, 1445–1452.
Koopman, T., Linde, C.C., Fourie, P.H. and McLeod, A. (2007) Population genetic structure of *Plasmopara viticola* in the Western Cape Province of South Africa. *Molecular Plant Pathology*, 8, 723–736.
Maddalena, G., Delmotte, F., Bianco, P.A., De Lorenzis, G. and Toffolatti, S.L. (2020) Genetic structure of Italian population of the grapevine downy mildew agent, *Plasmopara viticola*. *Annals of Applied Biology*, 176, 257–267.
Nita, M., Burrack, H., Pfeiffer, D., Mitchem, W., Lockwood, D., Bellinger, B. *et al.* (2016) *2016 Southeast Regional Bunch Grape Integrated Management Guide*. Bulletin 46. Athens, GA: College of Agricultural and Environmental Sciences, University of Georgia.
Peakall, R.O.D. and Smouse, P.E. (2006) GenAlEx 6: genetic analysis in Excel. Population genetic software for teaching and research. *Molecular Ecology Notes*, 6, 288–295.
Peakall, R. and Smouse, P.E. (2012) GenAlEx 6.5: genetic analysis in Excel. Population genetic software for teaching and research – an update. *Bioinformatics*, 28, 2537–2539.
Rossi, V., Caffi, T., Bugiani, R., Spanna, F. and Della Valle, D. (2008a) Estimating the germination dynamics of *Plasmopara viticola* oospores using hydro-thermal time. *Plant Pathology*, 57, 216–226.
Rossi, V., Caffi, T., Giosuè, S. and Bugiani, R. (2008b) A mechanistic model simulating primary infections of downy mildew in grapevine. *Ecological Modelling*, 212, 480–491.
Rouxel, M., Papura, D., Nogueira, M., Machefer, V., Dezette, D., Richard-Cervera, S. *et al.* (2012) Microsatellite markers for characterization of native and introduced populations of *Plasmopara viticola*, the causal agent of grapevine downy mildew. *Applied and Environmental Microbiology*, 78, 6337–6340.
Rouxel, M., Mestre, P., Baudoin, A., Carisse, O., Deliere, L., Ellis, M.A. *et al.* (2014) Geographic distribution of cryptic species of *Plasmopara viticola* causing downy mildew on wild and cultivated grape in eastern North America. *Phytopathology*, 104, 692–701.
Rumbou, A. and Gessler, C. (2004) Genetic dissection of *Plasmopara viticola* population from a Greek vineyard in two consecutive years. *European Journal of Plant Pathology*, 110, 379–392.
Rumbou, A. and Gessler, C. (2006) Particular structure of *Plasmopara viticola* populations evolved under Greek island conditions. *Phytopathology*, 96, 501–509.
Spring, O. and Zipper, R. (2000) Isolation of oospores of sunflower downy mildew, *Plasmopara halstedii*, and microscopical studies on oospore germination. *Journal of Phytopathology*, 148, 227–231.
Stenberg, P., Lundmark, M. and Saura, A. (2003) MLGsim: a program for detecting clones using a simulation approach. *Molecular Ecology Notes*, 3, 329–331.
Taylor, A.S., Knaus, B.J., Grünwald, N.J. and Burgess, T. (2019) Population genetic structure and cryptic species of *Plasmopara viticola* in Australia. *Phytopathology*, 109, 1975–1983.
Vercesi, A., Toffolatti, S.L., Zocchi, G., Guglielmann, R. and Ironi, L. (2010) A new approach to modelling the dynamics of oospore germination in *Plasmopara viticola*. *European Journal of Plant Pathology*, 128, 113–126.
Wolfe, K. and Stubbs, K. (2017) Grape farm gate value 2016. In: *2016 Georgia Farm Gate Value Report*. AR17-01. Athens, GA: College of Agricultural and Environmental Sciences, University of Georgia, pp. 51–52.
Wolfe, K., Kane, S.P. and Stubbs, K. (2013) Economic contribution of Georgia wineries and vineyards. *CAED Center Report CR-13-06*. College of Agricultural and Environmental Sciences, University of Georgia, Athens, GA.
Wong, F.P. and Wilcox, W.F. (2000) Distribution of baseline sensitivities to azoxystrobin among isolates of *Plasmopara viticola*. *Plant Disease*, 84, 275–281.
Wong, F.P. and Wilcox, W.F. (2001) Comparative physical modes of action of azoxystrobin, mancozeb, and metalaxyl against *Plasmopara viticola* (grapevine downy mildew). *Plant Disease*, 85, 649–656.
Yin, L., Zhang, Y.L. and Lu, J. (2010) Comparison of different DNA extraction methods for *Plasmopara viticola*. *Sino-overseas Grapevine and Wine*, 3, 4–7.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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